

EXP NO:4A

Secure Encryption and Decryption of Hospital Patient Records Using the DES Algorithm

DATE:

AIM:

ALGORITHM:

PROGRAM:

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
key = b'12345678'
cipher = DES.new(key, DES.MODE_ECB)
plaintext = input("Enter Plaintext: ")
ciphertext = cipher.encrypt(pad(plaintext.encode(),
DES.block_size))
print("Encrypted Data:", ciphertext.hex())
decipher = DES.new(key, DES.MODE_ECB)
```

```
decrypted = unpad(decipher.decrypt(ciphertext),  
DES.block_size)  
  
print("Decrypted Data:", decrypted.decode())
```

OUTPUT:

Enter Plaintext: HELLO

Encrypted Data:

3d7595a98bff809d

Decrypted Data:

HELLO

RESULT:

EXP NO:4B	Secure Online Banking Transaction Using the RSA Public-Key Cryptographic Algorithm
DATE:	

AIM:

ALGORITHM:

PROGRAM:

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a  
  
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:  
            return d
```

```
p = 17
q = 11
n = p * q
phi = (p - 1) * (q - 1)
e = 7
if gcd(e, phi) != 1:
    print("Choose another value for e")
else:
    d = mod_inverse(e, phi)
    print("Public Key:", (e, n))
    print("Private Key:", (d, n))
    message = int(input("Enter Message (<187): "))
    cipher = pow(message, e, n)
    print("Encrypted Message:", cipher)
    plain = pow(cipher, d, n)
    print("Decrypted Message:", plain)
```

OUTPUT:

```
Public Key: (7, 187)
Private Key: (23, 187)
Enter Message (<187): 44
Encrypted Message: 22
Decrypted Message: 44
```

RESULT:

EXP NO:4C

Software File Integrity Verification Using the MD5 Hashing Algorithm

DATE:

AIM:

ALGORITHM:

PROGRAM:

```
import hashlib
message=input("Enter Message: ")
hash_value=hashlib.md5(message.encode())
print("MD5 Digest:",hash_value.hexdigest())
```

OUTPUT:

Enter Message: HELLO

MD5 Digest: eb61eead90e3b899c6bcbe27ac581660

RESULT:

EXP NO: 4D	Scenario Cloud-Based Document Integrity Verification Using the SHA-1 Hashing Algorithm
DATE:	

AIM:

ALGORITHM:

PROGRAM:

```
import hashlib
message=input("Enter Message: ")
hash_value=hashlib.shal(message.encode())
print("SHA-1 Digest:",hash_value.hexdigest())
```

OUTPUT:

```
Enter Message: ASSET
SHA-1 Digest: a21b8b3c968deba7033aa40944f2456c879e997a
```

RESULT: